Optimising water delivery under constraints

CESQ Hackathon, 23-26 March 2026

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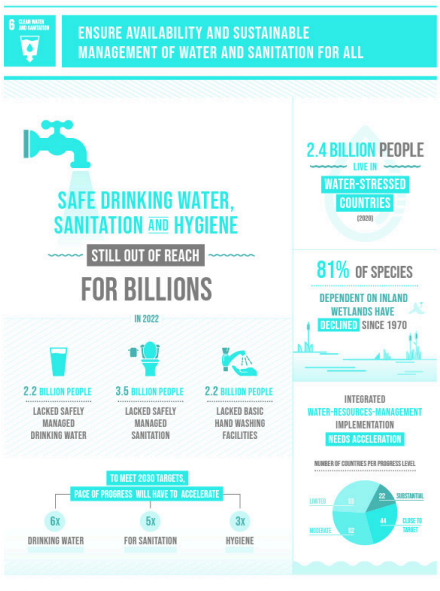
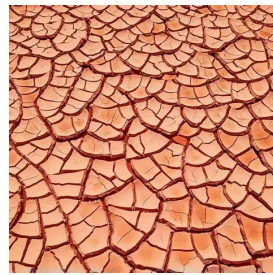
Context



One in four people does not have access to clean water



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Problem mapping

Typical water distribution networks are huge graphs

City size	Population	Nodes (junctions + tanks + reservoirs)	Edges (pipes + pumps + valves)
Small town	10k–50k	500 – 2,000	600 – 2,500
Medium city	50k–250k	2,000 – 10,000	2,500 – 12,000
Large city	250k–1M	10,000 – 50,000	12,000 – 60,000
Major metro	>1M	50,000 – 200,000+	60,000 – 250,000+

- Each demand region must have its requirements met
- Limited resources
- Infra: average node degree $\approx 2.2 - 2.6$
- Energetic: extraction/purification plants cannot run 24/7 & at the same time

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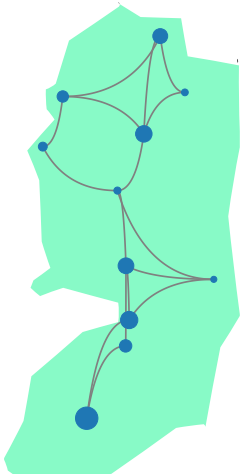
Dataset & problem formulation



Goal: maximise water delivery

$$\max_{\vec{x}} \left(\sum_{s \in \text{ Sources}} \left(\sum_{r \in \text{ Regions}} w_{sr} x_s x_r \right) - \text{constraints} \right)$$

Model and constraints:

- Each city has one water production source, on ($x_s = 1$) or off ($x_s = 0$)
- Not all can be on at the same time: $\sum_s x_s \leq S$
- Each source can deliver a maximum volume of water w_s: $\sum_{r \in \text{ Regions}} w_{sr} x_r \leq w_s x_s$
- Each consumption region should receive a minimum quantity of water w_r: $\sum_{s \in \text{ Sources}} w_{sr} x_s \geq w_r x_r$

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Proposed directions

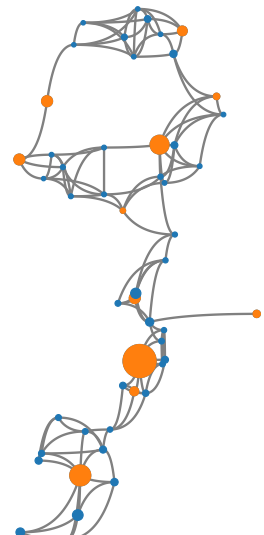
- Explore one or two quantum algorithms (VQE, QAOA, others)
- Constraints: how to include them in the optimisation?
- Solve simplified problems, then increment complexity

$$\max_{\vec{x}} \left(\sum_{\substack{s \in \text{ Sources} \\ r \in \text{ Regions}}} x_s w_{sr} - \mu \left(S - \sum_{s \in \text{ Sources}} x_s \right)^2 \right)$$

- Run your algorithms on a QPU
- Optimisation strategies: parameter transfer, scheduling
- Explore larger graphs

And most important:

Have fun!



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